MTH636 – Game Theory

Lecture 23: Strategies in Imperfect Information Games

1 Recap: The Formal Game

We continue with the extensive form game from Lecture 22. Consider the game tree rooted at n_0 (Player I), where Player I chooses A, B, or C:

- After A: reach n_1; after B: reach n_2. Player II has information set $h = \{n_1, n_2\}$, playing ℓ or g from each. Outcomes: n_4 with $(4, 1)$, n_5 with $(7, 2)$ from n_1; n_6 with $(6, 3)$, n_7 with $(4, 0)$ from n_2.
- After C: reach n_3. Nature (O) moves, $\frac{1}{4}$ to n_7 (Player II moves L/R) and $\frac{3}{4}$ to n_8 (Player I moves a/b). After Player II's move at n_7: n_{10} with $(0, 0)$ and n_{11} with $(8, 4)$. After Player I's move at n_8: n_{12} with $(8, 8)$ and n_{13} with $(0, 8)$.

The information sets are:

$$V_1 = \{n_0, n_8\}, \quad V_2 = \{n_1, n_2, n_8\}$$

$$H_1 = \Big\{\{n_0\},\, \{n_8\}\Big\}, \quad H_2 = \Big\{\{n_1, n_2\},\, n_8\Big\}$$

1.1 Actions at an Information Set

For the information set $h = \{n_1, n_2\}$:

$$A(h) = \Big\{\{\{n_1, n_4\},\, \{n_2, n_6\}\}, \quad \{\{n_1, n_5\},\, \{n_2, n_7\}\}\Big\}$$

That is, the two actions at h are ℓ (= choosing the left branch from whichever node in h the player is at) and g (= choosing the right branch), where each action bundles together one edge from each node in h.

2 Pure Strategies

Definition: Pure Strategy

A **pure strategy** for player i, denoted s_i, is a mapping

$$s_i : H_i \longrightarrow \bigcup_{h \in H_i} A(h) \quad \text{such that} \quad s_i(h) \in A(h) \quad \forall\, h \in H_i.$$

That is, a pure strategy assigns an action to every information set of player i, even those that may not be reached under the play induced by s_i.

The **pure strategy set** S_i for player i is the set of all such mappings.

The **pure strategy profile space** is:

$$S = S_1 \times S_2 \times \cdots \times S_n.$$

The **utilities at strategy profiles** are defined as the **expected utilities** (since there may be chance nodes).

3 Mixed Strategies

Definition: Mixed Strategy

Mixed strategies are probability distributions over pure strategies. For player i, the set of all mixed strategies is denoted Σ_i.

The **strategy profile** for mixed strategies is $\sigma = (\sigma_1, \ldots, \sigma_n)$.

Note

Utilities are defined as convex combinations of utilities at pure strategy profiles.

3.1 Example

Consider a two-player game tree: Player I (n_0) chooses T_1 or B_1 (reaching n_1 or n_2). Player II (n_2) has actions L or R (reaching n_2's subtree). Player I also has a second information set containing $\{n_3, n_4\}$ where she plays T_2 or B_2. Player II has another information set $\{n_3, n_4, n_7\}$ where she plays β_2.

The pure strategy set of Player I is:

$$S_1 = \{T_1T_2,\ T_1B_2,\ B_1T_2,\ B_1B_2\}$$

The mixed strategy set of Player I is:

$$\Sigma_1 = \Delta_4 = \left\{ (n_1, \ldots, n_4) \;\middle|\; n_i \geq 0,\ \sum_{i=1}^{4} n_i = 1 \right\}$$

The set of **behaviour strategies** B_1 for Player I (explained below) is:

$$B_1 = \{(n, 1-n),\ (y, 1-y) \mid 0 \leq n \leq 1,\ 0 \leq y \leq 1\}$$

This is the set of behaviour strategies – one independent randomisation per information set.

4 Behaviour Strategies

Definition: Behaviour Strategy

A **behaviour strategy** for player i is a mapping

$$b_i : H_i \longrightarrow \bigcup_{h \in H_i} \Delta\Big(A(h)\Big) \quad \text{such that} \quad b_i(h) \in \Delta\Big(A(h)\Big) \quad \forall\, h \in H_i.$$

Here $\Delta(S)$ denotes a probability distribution over a set S.

Thus a behaviour strategy assigns to each information set an *independent* probability distribution over the available actions at that information set.

- For a behaviour strategy b_i, the probability that b_i selects action a at information set h is denoted $b_i(a, h)$.
- A **behaviour strategy profile** is $b = (b_1, \ldots, b_n)$.
- With a slight abuse of notation, we write $\sigma = (\sigma_i)_{i \in N}$ for a strategy profile where σ_i can be either a mixed or behaviour strategy for player i.

5 Reaching Probabilities

Definition: Reaching Probability

For a mixed/behaviour strategy profile $\sigma = (\sigma_i)_{i \in N}$, and a node $n \in V$, we denote by $\rho(n; \sigma)$ the **probability that node n will be reached** when the underlying strategy profile is σ.

5.1 Example Computation

Consider the following game tree: Player I (n_0) chooses A or B (reaching n_1 or n_2). Player II has information set $\{n_1\}$ (singleton), and plays L or R (reaching n_3 or n_4). Player I has a second information set $\{n_3, n_4\}$ where she plays ℓ or r (reaching n_5, n_6, n_7, n_8).

Suppose:

$$\sigma_1 = \left[\tfrac{2}{10}(A\ell),\ \tfrac{1}{10}(Ar),\ \tfrac{4}{10}(B\ell),\ \tfrac{2}{10}(Br)\right] \quad \text{(mixed strategy for Player 1)}$$

$$\sigma_2 = \left[\tfrac{1}{2}(L),\ \tfrac{1}{2}(R)\right] \quad \text{(mixed strategy for Player 2)}$$

Computing $\rho(n_5; \sigma)$: n_5 will be reached when Player 1 plays b:

$$\rho(n_5; \sigma) = \tfrac{2}{10} \times \tfrac{1}{2} = \tfrac{2}{20} = \tfrac{1}{10}$$

Computing $\rho(n_2; \sigma)$: n_2 will be reached if Player 1 plays either $A\ell$ or Ar, *and* Player 2 plays L. Also the prob is:

$$\left[\tfrac{2}{10} + \tfrac{1}{10}\right] \times \tfrac{1}{2} = \tfrac{4}{20}$$

6 Equivalence of Mixed and Behaviour Strategies

Definition: Equivalence of Mixed and Behaviour Strategies

A **mixed strategy** σ_i and a **behaviour strategy** b_i of player i are **equivalent** if for every mixed/behaviour strategy vector σ_{-i} of the other players and every vertex $n \in V$,

$$\rho\Big(n;\ (\sigma_i, \sigma_{-i})\Big) \ = \ \rho\Big(n;\ (b_i, \sigma_{-i})\Big).$$

That is, two strategies (one mixed, one behaviour) are equivalent if they induce identical probability distributions over the nodes of the game tree, regardless of what the other players do.

Key Takeaway

Key Insight. This equivalence definition says that σ_i and b_i produce the same reaching probabilities for *every* node in the game tree, for *every* possible strategy profile of the opponents. Kuhn's theorem then states that under **perfect recall**, every mixed strategy has an equivalent behaviour strategy and vice versa.